METHODOLOGY REPORT

An Early-Warning System for US Bank Failure

A distress screen built entirely from FDIC call reports, developed on the 2008-12 failure wave and tested on 2013-2023 — including the failures it could not see.

Panel	588,177 bank-quarters, 10,459 institutions, 2003-2023, quarterly
Events	563 dated bank failures matched to pre-failure call reports
Development	529 failures before 2013 — the 2008-12 wave
Held-out test	63 failures 2013-2023, a different regime
Headline	54 of 63 caught (86%), median lead 15.8 months, 24.8x precision lift
Counter-headline	**4% of failed-bank assets**, and no better than a one-line benchmark

Academic work on public data. Not investment advice, not a supervisory tool.

CONTENTS

What is in this report

Part	Contents
I — Problem and data	Why bank failure is predictable in principle, and what the FDIC publishes.
II — Method	Nine ratios, peer-relative percentiles, a weighted index, and an alert rule with two legs.
III — Development	Component and threshold selection on the 2008-12 wave, including a flaw in the selection rule.
IV — Held-out results	Performance on 2013-2023, a regime the model never saw.
V — What the model cannot see	The 2023 failures, and why a credit model is blind to a duration run.
VI — Limitations	What the evidence does not support.

PART I

Problem and Data

A bank fails on one day. The condition that causes it develops over years and is filed quarterly with the regulator.

1. Why this is a tractable problem

US banks file a Consolidated Report of Condition and Income — the call report — every quarter. Capital, earnings, loan quality and funding are all reported on a common schedule and a common definition, for every insured institution. Separately, the FDIC publishes every failure since 1934 with its date. The two join on a single identifier.

This makes bank failure one of the few credit problems with a genuinely large, clean, dated event set: **563 failures with pre-failure financials** inside this window. Most corporate-distress work has to make do with dozens.

Cole and White [5] find that the 2008-09 failures were explained well by ordinary call-report ratios — commercial real-estate concentration and thin capital. That is the hypothesis this system operationalises and tests out of sample.

2. The data

Source	Content	Scale
FDIC *financials* endpoint	Quarterly call reports: assets, equity, deposits, net income, interest income and expense, non-performing assets, loss reserves	588,177 bank-quarters
FDIC *failures* endpoint	Every failure with date, resolution type and assets at failure	4,117 records; 563 in window

Both are free and require no authentication. Nothing else is used — no market prices, no ratings, no supervisory data.

No survivorship bias. Failed banks remain in the panel with the call reports they filed before failing. This is the defect that makes most commercial financial databases unusable for this question: they drop institutions once those institutions cease to exist, which removes precisely the observations the exercise needs.

Point-in-time. A call report for a quarter end is treated as public 45 days later. No result uses information that was unavailable on the date the prediction is attributed to.

PART II

Method

Nine ratios, each converted to a rank against comparable banks in the same quarter, combined into one index, with an alert rule that fires on level or on movement.

3. Components

Family	Component	Definition	Weight
Capital	Capital buffer	equity / assets	0.20
	Capital erosion	8-quarter change in equity, scaled by assets	0.14
	Leverage	(assets – equity) / equity	0.08
Asset quality	Non-performing ratio	non-performing assets / assets	0.18
	Reserve shortfall	non-performing assets not covered by loss reserves	0.08
Earnings	Profitability	return on assets	0.14
	Earnings trend	12-quarter change in ROA	0.06
Funding	Deposit contraction	4-quarter fall in deposits	0.06
	Interest burden	interest expense / interest income	0.06

Weights are set by economic reasoning, not fitted. With 529 development events they could be fitted; they are not, so that the held-out result measures the design rather than the fitting.

4. From ratio to score

```
1. orient    every ratio so that a larger value means more risk
2. rank      within (quarter x size tercile), as a percentile 0-100
3. anchor    capital components also ranked against all banks that quarter,
             and the two ranks averaged
4. override  equity <= 0 forces the capital components to 100
5. weight    index = sum(percentile x weight) / sum(weight available)
             null if fewer than 60% of components are available
```

Ranking rather than thresholding makes the index self-calibrating: in a quarter when every bank's capital is falling, the ranking still identifies who is weakest. Size terciles keep a community bank from being judged against a money centre. The second, whole-population anchor on capital exists because a tercile that is uniformly thin would otherwise look safe by its own standards.

5. Alert rule

```
alert  =  index >= 85
       OR ( index >= 65  AND  index - median(own last 12 quarters) >= +20 )
```

Two legs, because they answer different questions. A high level identifies *which* banks are weak; a sharp move against a bank's own history identifies *when* something changed. A bank that has scored badly for five years is a standing concern, not a warning. Thresholds are derived in Part III and never adjusted afterwards.

PART III

Development

Components and thresholds are chosen using only failures before 2013, then frozen. The held-out period is a different regime, not merely later dates.

6. Component selection

Each candidate is scored by its ability to separate pre-failure bank-quarters from the rest, on development data alone. All nine clear a coin flip, so none is dropped.

Component	AUC, development data
Non-performing ratio	**0.875**
Reserve shortfall	**0.856**
Profitability	0.830
Leverage	0.815
Capital buffer	0.814
Earnings trend	0.783
Interest burden	0.779
Capital erosion	0.768
Deposit contraction	0.580

Asset quality dominates, which is what a credit-driven failure wave should look like. **Deposit contraction is the weakest of the nine.** Part V shows why that ordering, learned from 2008-12, is the system's central weakness.

7. A flaw in the threshold rule, caught and corrected

Thresholds were first chosen by maximising development recall, with watchlist size only as a tie-break. With 529 events that rule simply loosens the threshold until nearly everything is flagged. It selected a watchlist of **18% of the banking system** — a list no analyst would read, and the exact failure this kind of model is usually criticised for.

The rule was replaced, before re-running, with: **maximise recall subject to a watchlist no larger than 5% of banks**. The capacity of the person reading the list is a modelling constraint and belongs in the objective, not in the commentary afterwards.

Selection rule	Thresholds	Held-out watchlist	Held-out precision
Maximise recall	70 / 50 / +10	18% of banks	0.74% (4.7x lift)
Recall subject to 5% cap	**85 / 65 / +20**	**2.8% of banks**	**3.93% (24.8x lift)**

On development data the chosen rule catches 504 of 529 failures with 381 banks on the list per quarter.

PART IV

Held-Out Results

2013-2023: the post-crisis tail. The model was developed on a credit wave and is tested on a period that contains almost none.

8. Headline

Measure	Value
Failures alerted within 24 months	**54 of 63 (86%)**
Median lead time	15.8 months (quartiles 8.2 / 22.2)
Watchlist	159 of 5,675 banks per quarter (2.8%)
Precision	3.93% against a 0.16% base rate — **24.8x lift**
Permutation test	observed 55, null mean 4.6, **$p < 0.001$**

The permutation null preserves failure dates and size terciles and permutes only *which* bank failed, over 1,000 draws. Chance would have caught about five.

9. By year

Year	Failures	Alerted
2013	24	24
2014	18	16
2015	8	8
2016	5	5
2017	8	7
2019	4	3
2020	4	4
2023	**5**	**1**

Seven consecutive years at or near complete recall, then a collapse. The remainder of this report is about that last row.

10. What the asset-quality components contributed

Model	Caught	Median lead	Precision	AUC
Six components (capital, earnings, interest)	55 / 68	14.4 months	2.04%	0.895
Nine, adding asset quality and funding	**55 / 68**	14.5 months	**3.93%**	0.894

Identical recall, lead and AUC — and half the false alarms. Asset quality is the strongest standalone signal in the system, yet it changes nothing about *which* banks are caught, because capital and earnings already identify them. What it adds is discrimination among the many banks that look weak, which halves the list needed to catch the same failures.

PART V

What the Model Cannot See

In March 2023 three of the four largest bank failures in US history occurred inside the held-out window. The model flagged none of them.

11. Recall by count and by assets

Period	Failures caught	Failed assets caught
2013-2022	53 of 58 (91%)	**$20.1bn of $20.4bn (99%)**
2023	**1 of 5**	**$0.1bn of $532.2bn (0.02%)**
Full held-out period	54 of 63 (86%)	**4%**

By count the system looks strong. By assets it caught 4% of what failed. Those two numbers describe the same model on the same data, and the gap between them is the most important result in this report.

12. What Silicon Valley Bank's last call report said

SVB filed for the quarter ending 31 December 2022, public in February 2023. It failed on 10 March 2023. The figures, in $ thousands:

```
assets              209,026,000       equity / assets      7.39%
equity               15,456,000       return on assets     0.96%
deposits            175,378,000       non-performing      0.066%
                (191,431,000 a year earlier,  -8.4%)
```

On every measure this system reads, SVB was a well-capitalised, profitable bank with almost no bad loans. Its non-performing ratio sat in the best quartile of its peer group. The index scored it **59.0** against an alert threshold of 85, or 65 with a sharp move.

Bank	Last index	Capital	Non-performing	Profitability	Deposit contraction
Silicon Valley Bank	59.0	75	25	70	**94**
Signature Bank	61.9	77	53	51	**98**
First Republic Bank	48.6	63	21	78	19

Component percentiles against peers, on the last call report published before failure. 100 is worst in the peer group.

One component did fire. Deposit contraction put SVB at the 94th percentile and Signature at the 98th — the deposit flight was visible in the filings. But deposit contraction was the weakest component in development (AUC 0.580, Part III) and carries a 6% weight, so it could not move a composite dominated by capital and asset quality.

13. It does not beat a one-line benchmark

The Texas Ratio — non-performing assets divided by tangible equity plus loss reserves — is the standard bank-distress screen, computable in one line. Any claim that a nine-component index is worth building has to survive comparison with it. At a matched 2.8% watchlist rate on the held-out period:

Signal	AUC	Failures caught	Precision
Nine-component index	0.894	56 of 63	3.98%
Texas Ratio	0.869	56 of 63	4.02%
Equity / assets alone	**0.913**	**56 of 63**	3.57%

All three catch the same 56 failures. The composite adds no recall over a single ratio, and on this period a bare capital ratio has the higher AUC. What the composite provides is a decomposable score with an explicit reason attached to each alert — interpretability, not accuracy.

This also qualifies a claim made elsewhere in this project. Over the full 2003-2023 period the composite does beat a leverage ratio (0.861 against 0.837). On the held-out period the ranking reverses. The advantage is not robust to the period chosen, and should not be presented as though it were.

14. Why a credit model is blind to a duration run

The 2008-12 failures, on which this model was built, were credit events: banks made bad loans, the loans went bad, reserves and then capital absorbed the losses, and the bank closed. Every stage of that sequence is visible in the ratios above, which is why the model achieves 99% asset recall in 2013-2022.

The 2023 failures were not credit events. Jiang and co-authors [6] describe the mechanism: banks held long-dated securities bought at low rates, rate rises cut their market value, and the resulting unrealised losses were large relative to equity but were **not recognised in the reported figures** for securities held to maturity. A concentrated, uninsured depositor base then withdrew faster than the bank could sell assets without crystallising those losses.

None of that appears in the nine components. The fields that would capture it are available in call reports and were simply not pulled here:

- **Unrealised securities losses** (accumulated other comprehensive income, and the amortised-cost versus fair-value gap on held-to-maturity portfolios).
- **Held-to-maturity share of assets** — how much of the balance sheet is carried at a value the market no longer agrees with.
- **Uninsured deposit share** — how much of the funding can leave in a day.
- **Securities duration** — how much value a given rate move destroys.

These were not added and the model was not re-run. Adding features after seeing which failures were missed, and then reporting improved recall on those same failures, would be fitting to the test set. The honest statement is that this system detects credit-driven failure and is structurally blind to interest-rate-driven runs, and that the fix is specified above but untested.

The general lesson is not about banks. A model trained on one failure mechanism measures that mechanism. It will score well on a held-out sample drawn from the same mechanism — seven straight years here — and fail completely the first time the mechanism changes. Out-of-sample testing does not protect against this, because the new regime is not in any sample yet.

PART VI

Limitations

Read this before quoting any figure above.

15. Scope of the claim

- **The 4% asset-recall figure is the headline, not the 86%.** The system is demonstrated on small and mid-sized credit-driven failures. It has no demonstrated ability on large institutions, and one clear demonstration of failure on them.
- **The held-out period is thin.** 63 failures, and 2014-2022 contributes fewer than 35. Seven years of near-complete recall rests on a small annual count.
- **Development is one crisis.** 529 of the events come from a single 2008-12 wave with a common cause. The model learned that cause well.
- **The index is a rank, not a probability.** A score of 85 means worse than 85% of comparable banks that quarter. It is not a probability of failure, and most banks above the threshold do not fail: precision is 3.93%.

16. Data and construction

- **Call-report figures are as filed.** Reported non-performing assets depend on a bank's own classification decisions, and the 2017 failure missed here (Washington Federal Bank for Savings) was a fraud case in which the filings were false.
- **Securities and interest-rate risk are entirely absent** from the feature set, as Part V describes.
- **Failures after 2023 cannot be evaluated** because the call-report panel ends at 2023Q4, so the four failures in 2025-2026 are excluded rather than counted as misses.
- **Size terciles are recomputed each quarter**, so a growing bank can change peer group, and therefore score, without changing itself.
- **Weights are judgement.** They were not fitted, which protects the held-out test, but it also means no claim is made that they are optimal.

17. Reproducing this

```
pip install pandas numpy

python3 fdic_fetch.py      # FDIC failures + call reports -> fdic_panel.csv
python3 fdic_model.py      # components, thresholds, held-out test

# roughly 90 API calls, no authentication, about 15 minutes
```

Every figure in this report is produced by those two scripts.

References

[1] R. C. Merton, “On the pricing of corporate debt: The risk structure of interest rates,” *J. Finance*, vol. 29, no. 2, pp. 449–470, 1974.

[2] E. I. Altman, “Financial ratios, discriminant analysis and the prediction of corporate bankruptcy,” *J. Finance*, vol. 23, no. 4, pp. 589–609, 1968.

[3] Federal Deposit Insurance Corporation, “BankFind Suite: API for developers,” FDIC. [Online]. Available: https://api.fdic.gov

[4] Federal Financial Institutions Examination Council, “Consolidated Reports of Condition and Income (Call Reports),” FFIEC.

[5] R. A. Cole and L. J. White, “Déjà vu all over again: The causes of U.S. commercial bank failures this time around,” *J. Financial Services Research*, vol. 42, pp. 5–29, 2012.

[6] E. X. Jiang, G. Matvos, T. Piskorski and A. Seru, “Monetary tightening and U.S. bank fragility in 2023: Mark-to-market losses and uninsured depositor runs?” NBER Working Paper 31048, 2023.